School Worksheet

# Class 12 Mathematics, Chapter 11: Three Dimensional Geometry

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1

8 marks

**1** MCQ — 1 mark

The direction cosines of a line equally inclined to the coordinate axes are:

**(A)** $\pm 1, \pm 1, \pm 1$

**(B)** $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$

**(C)** $\pm \frac{1}{3}, \pm \frac{1}{3}, \pm \frac{1}{3}$

**(D)** $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}$

**2** MCQ — 1 mark

If a line makes angles $90^\circ$, $135^\circ$ and $45^\circ$ with the x, y and z axes respectively, its direction cosines are:

**(A)** $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

**(B)** $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

**(C)** $1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

**(D)** $0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

**3** MCQ — 1 mark

The Cartesian equation of the line through $(5, -2, 4)$ and parallel to $3\hat{i} + 2\hat{j} - 8\hat{k}$ is:

**(A)** $\frac{x+5}{3} = \frac{y-2}{2} = \frac{z+4}{-8}$

**(B)** $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$

**(C)** $\frac{x-3}{5} = \frac{y-2}{-2} = \frac{z+8}{4}$

**(D)** $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{8}$

| Q No. | Question | Marks |
|---|---|---|
| 4 | MCQ<br>The angle between the lines $\frac{x-2}{2}=\frac{y-1}{5}=\frac{z+3}{-3}$ and $\frac{x+2}{-1}=\frac{y-4}{8}=\frac{z-5}{4}$ is:<br>**(A)** $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$<br>**(B)** $\cos^{-1}\left(\frac{26}{9}\right)$<br>**(C)** $\cos^{-1}\left(\frac{13}{9\sqrt{38}}\right)$<br>**(D)** $\frac{\pi}{2}$ | 1 mark |
| 5 | MCQ<br>The value of $p$ for which the lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are perpendicular is:<br>**(A)** $\frac{70}{11}$<br>**(B)** $\frac{11}{70}$<br>**(C)** $-\frac{70}{11}$<br>**(D)** $7$ | 1 mark |
| 6 | Assertion-Reason<br>**Assertion (A):** The line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ passes through the origin. **Reason (R):** Any line of the form $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ passes through the origin, since $(0,0,0)$ satisfies it.<br>**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).<br>**(C)** Assertion (A) is true, but Reason (R) is false.<br>**(D)** Assertion (A) is false, but Reason (R) is true. | 1 mark |
| 7 | Short answer<br>Find the vector and Cartesian equations of the line passing through $A(1,2,-1)$ and $B(2,1,1)$. | 2 marks |

## Level 2

16 marks

| Q No. | Question | Marks |
|---|---|---|
| 8 | Short answer<br>Show that the lines $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ are perpendicular to each other. | 2 marks |

| Q No. | Question | Marks |
|---|---|---|
| 9 | Short answer<br>Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$. | 3 marks |
| 10 | Short answer<br>Find the distance between the parallel lines $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$. | 3 marks |
| 11 | Short answer<br>Find the angle between the lines $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$. | 2 marks |
| 12 | Short answer<br>Find $k$ so that the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$ are perpendicular. | 2 marks |
| 13 | Short answer<br>Find the direction cosines of the line whose Cartesian equations are $5x - 3 = 15y + 7 = 3 - 10z$. | 2 marks |
| 14 | Short answer<br>Find the Cartesian equation of the line through $(-2, 4, -5)$ and parallel to $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$. | 2 marks |
| **Level 3** | | **27 marks** |
| 15 | Long answer<br>Find the coordinates of the foot of the perpendicular drawn from $P(1, 2, 3)$ to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Also find the length of the perpendicular. | 5 marks |
| 16 | Long answer<br>Find the image $A'$ of the point $A(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Also, find the equation of the line joining $A$ and $A'$. | 5 marks |
| 17 | Long answer<br>Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Find their point of intersection. | 5 marks |
| 18 | Long answer<br>Find a point $P$ on the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ such that its distance from the point $Q(2, 4, -1)$ is 7 units. Also, find the equation of the line joining $P$ and $Q$. | 5 marks |
| 19 | Short answer<br>Find the vector equation of the line passing through $(1, 2, -4)$ and perpendicular to both the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$. | 3 marks |

| Q No. | Question | Marks |
|---|---|---|
| 20 | Case study<br>**Case Study:** Two aircraft fly along straight paths given by $L_1 : \vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$. Air-traffic control needs to know how close the paths come. | 4 marks |
| | **(i)** Write the direction ratios of $L_1$ and $L_2$. | 1 mark |
| | **(ii)** Are the paths parallel? Justify. | 1 mark |
| | **(iii)** Find the shortest distance between the two paths. | 2 marks |

# Answer key

Chapter 11: Three Dimensional Geometry, School Worksheet. Full step-wise solutions are in the solutions sheet.

## Level 1

1 (B) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

2 (B) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

3 (B) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$

4 (A) $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

5 (A) $\frac{70}{11}$

6 (A) Both A and R are true and R is the correct explanation of A.

7 $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k})$; $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+1}{2}$

## Level 2

8 $7(1) + (-5)(2) + 1(3) = 0$, so the lines are perpendicular.

9 Shortest distance $= \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ units

10 Distance $= \frac{\sqrt{293}}{7}$ units

11 $\theta = \cos^{-1}\left(\frac{19}{21}\right)$

12 $k = -\frac{10}{7}$

13 $\frac{6}{7}, \frac{2}{7}, -\frac{3}{7}$

14 $\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$

## Level 3

15 Foot of perpendicular $M(3, 5, 9)$; length of perpendicular $= 7$ units

16 Image $A'(1, 0, 7)$; line $AA'$: $\vec{r} = (\hat{i} + 6\hat{j} + 3\hat{k}) + \mu(-3\hat{j} + 2\hat{k})$, i.e. $\frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$

17 The lines intersect at $\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$.

18 $P(-4, 1, -3)$; line $PQ$: $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

19 $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

20 (i) $2, -1, 1$ and $3, -5, 2$; (ii) No, DRs are not proportional; (iii) $\frac{10}{\sqrt{59}}$ units